Francesco Granella

Environmental economist

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October, 2026

Summary

I am an environmental economist focusing on the economic implications of environmental stressors. My research agenda is structured on two complementary pillars: advancing the empirical estimates of environmental impacts, and improving our understanding of climate change damages. My methodological toolkit spans from causal inference to machine learning to integrated assessment models.

Positions & affiliations

RFF-CMCC European Institute on Economics and the Environment — **Jan 2025 – Present**
Jr Scientist — Milan, Italy

Bocconi University — **Sep 2023 – Present**
Academic Fellow — Milan, Italy

RFF-CMCC European Institute on Economics and the Environment — **Jan 2023 – Dec 2024**
Postdoctoral fellow — Milan, Italy

University of California, Davis — **2021 – 2022**
Visiting Scholar — Davis, CA, USA

COVID Crisis Lab, Bocconi University — **2020– 2023**
Affiliated student — Milan, Italy

RFF-CMCC European Institute on Economics and the Environment — **2018 – 2022**
Post-Degree Junior Researcher — Milan, Italy

GREEN Research Center, Bocconi University — **2018– 2023**
Affiliated student — Milan, Italy

Fondazione Eni Enrico Mattei — **Oct 2016 – Jul 2018**
Post-Degree Junior Researcher — Milan, Italy

Education

Bocconi University — **2017 – Jan 2023**
PhD in Public Policy & Administration — Milan, Italy

Bocconi University — **2013 – Apr 2016**
MSc in Economic and Social Sciences — Milan, Italy

Sciences Po Paris — **2014**
Exchange program — Paris, France

Bocconi University — **2010 – 2013**
BSc in Int. Economics, Management and Finance — Milan, Italy

Insper — **2012**
Exchange program — São Paulo, Brazil

Publications

- Bernardo A. Bastien-Olvera, Octavio Aburto-Oropeza, Luke M. Brander, William W. L. Cheung, Johannes Emmerling, Christopher M. Free, Francesco Granella, Massimo Tavoni, Jasper Verschuur and Katharine Ricke (2026) "Accounting for ocean impacts nearly doubles the social cost of carbon", *Nature Climate Change*. [Link]

- Stefania Renna, Francesco Granella, Lara Aleluia Reis,and Paulina Antipa (2025) "CLAQC v1. 0 - Country Level Air Quality Calculator: an empirical modeling approach", *Geoscientific Model Development* [Link][Replication]

- Bernardo Bastien-Olvera, Mark Conte, Xiaoli Dong, Tania Briceno, David Batker, Johannes Emmerling, Massimo Tavoni, Francesco Granella, and Frances Moore (2024) "Unequal climate impacts on global values of natural capital", *Nature* [Link]

- Stefania Renna, Francesco Granella, Jacopo Lunghi, Michele Malpede, and Damiano Di Simine (2024) "Impacts of agriculture on PM10 pollution and human health in the Lombardy region in Italy", *Frontiers in Environmental Science* [Link]

- Martino Gilli, Matteo Calcaterra, Johannes Emmerling, and Francesco Granella (2024) "Climate Change Impacts on the Within-Country Income Distributions", *Journal of Environmental Economics and Management* [Link][Replication package]

- Francesco Granella, Stefania Renna, Lara Aleluia Reis (2024) "The formation of secondary inorganic aerosols: A data-driven investigation of Lombardy's secondary inorganic aerosol problem", *Atmospheric Environment*. [Link]

- Bernardo A. Bastien-Olvera, Francesco Granella, and Frances Moore (2022). "Persistent effect of temperature on GDP identified from lower frequency temperature variability", *Environmental Research Letters*. [Link][Replication package]

- Francesco Granella, Lara Aleluia Reis, Valentina Bosetti, and Massimo Tavoni (2020) "COVID-19 lockdown only partially alleviates health impacts of air pollution in Northern Italy", *Environmental Research Letters*. [Link][Replication package]

Working papers

- *The effect of air purifiers in schools.* With Jacopo Bonan, Stefania Renna, Alejandro Sarmiento, and Massimo Tavoni. [Link].
- *The Impact of Weather Extremes on Firm-Level Productivity.* With Johannes Emmerling, Diego Paroli, Alessandro Taberna.
- *Global economic connections increase returns to mitigation efforts.* With Pietro Andreoini. Draft available soon.
- *Estimating Climate Damages: Labor Availability and the Timing of Weather Shocks.* With François Bareille.
- *Understanding the representation of climate impacts and adaptation in Integrated Assessment Models* With Mathijs Harmsen, Johannes Emmerling, Laurent Drouet, et al.
- *Heterogeneous Effects of Air Pollution on Physical Tasks: Evidence from Amateur Track and Field.* [Link]
- *Campaign Money for Nothing? Understanding the Consequences of a Ban on Corporate Contributions: Evidence from Brazil* With Alexandros Cavgias [Link]

Teaching

Instructor of Record

- *Social Transformations*, 20 hours, at Bocconi University. Student evaluation: 9.07/10 (Ug, Fall 2025)

Visiting Scholar & Guest Lectures

- *Economics and Econometrics of Air pollution* at Politecnico di Milano (MSc, 2025)
- *Synthetic Control Methods and ML for Impact Evaluation* at University of Brescia (MSc, 2024)
- *Climate Change Impacts* at Bocconi University (Ug, 2023)
- *Data Science for Policy Evaluation* at University of Aosta (Ug, 2020)

Teaching assistant

- *Macroeconomics and the World Economy* (Ug), *2025*
- *Climate Change Economics* (Ug), *2021, 2022, 2023, 2024.*
- *Applications for Management* (Ug), *2019, 2020.*
- *Political Science and Comparative Politics* (Ug), *2019, 2020.*
- *Comparative Politics: Institutions and Institutional Development* (MSc), *2020.*

Supervision

- Carlotta Patroni, "*Biodiversity and climate change: the role of natural capital investments in a coupled economy-climate model*", MSc in Mathematical Engineering @ Politecnico di Milano, Co-supervision with Massimo Tavoni, 2024.

Conferences & seminars

Selected talks

2026: 3CMFI Frankfurt, Asian Development Bank (*Invited Seminar*), WCERE Lisbon, AIBE Milan (*Invited discussant*), University of Galway

2025: AgroParisTech–INRAE–Université Paris-Saclay *(Invited Seminar)*, EAERE Annual Meeting, IAMC Annual Meeting, IAERE Annual Meeting

2023: FAERE Annual Meeting

2022: Stockholm School of Economics Economics of Air Pollution Workshop

2021: AERE @ SEA, IAERE Annual Meeting, EAERE Annual Meeting

Organization

2023-Present: Seminar Series, RFF-CMCC European Institute on Economics and the Environment.

2025-Present: Climate Impacts Working Group, RFF-CMCC European Institute on Economics and the Environment.

2023: EIEE Summer School on Integrated Assessment Models.

Service

Referee for: *Nature Climate Change, Nature Communications, iScience, Environmental Research Letters, Climatic Change, Global Environmental Change, Environmental Research Communications, Environmental Science and Pollution Research.*

Project management

- Task leader, 101081369 Horizon Europe programme ("SPARCCLE"). 2023 - Present
- Task leader, 101081604 Horizon Europe programme ("PRISMA"). 2023 - Present

Grants & Funds

- Co-winner, *INHALE: Impact on humaN Health of Agriculture and Livestock Emissions* , WP1. PI: Lara Aleluia Reis. (€125,000)
- GEMCLIME (Global Excellence in Modelling of Climate and Energy) project - Secondment research allowance (€14,000)
- Invernizzi PhD Financial Aid grant (€22,000)

Awards

- 2023-2024 Bocconi Teaching Award, a university-wide award for excellence in teaching with a €2500 prize.
- EAERE 2021 Conference Participation Grant.
- PhD Fellowship, Bocconi University. 4-year full scholarship for the PhD in Public Policy and Administration.
- *Cum laude* honours, MSc in Economic and Social Sciences. University.

Computer skills

Python, Stata, R, GAMS, Git, LaTeX, HPC, Agentic AI.

Consultancy

2026: Assessment of ocean-related climate impacts for the Asian Development Bank
2020/22: Air pollution emissions-concentrations functions using machine learning, for the World Bank.
2019: Assessment of transport demand using mobile phone data for Trenitalia SpA.
2019: Assessment of market potential for firms in Lombardy and simulations of the impact of changes in transport costs, for the Union of the Chambers of Commerce of Lombardy.

Non-academic Experience

OECD Development Centre — **Jul – Oct 2015**
Jr Policy Analyst — Paris, France

OECD Development Centre — **Jan – Jul 2015**
Trainee — Paris, France

Languages

English: Full professional proficiency
Portuguese: Full professional proficiency
French: Limited professional proficiency
Italian: Mother tongue

References

Valentina Bosetti
Full Professor, Bocconi University
Department of Economics
valentina.bosetti@unibocconi.it

Massimo Tavoni
Full Professor, Politecnico di Milano
School of Management
Director, RFF-CMCC European Institute on Economics and the Environment
massimo.tavoni@cmcc.it

Bernardo Bastien-Olvera
Assistant Professor, Universidad Nacional Autónoma de México,
Atmospheric Sciences Department
bbastien@atmosfera.unam.mx

Johannes Emmerling
Senior Scientist, RFF-CMCC European Institute on Economics and the Environment
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